EFY TESTED S.C. Dwivedi

EMERGENCY LIGHT Cum ADVERTISING DISPLAY BOARD

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The electronics circuit presented here serves as both an emergency light as well as an advertising display board. It relies on a 12-volt, 1.3Ah DC battery for power backup, ensuring more than two hours of illumination after a mains power outage. Fig. 1 displays the author's fully assembled prototype of the emergency light cum advertising display board featuring the alphabets 'V' and 'J.'

Circuit and its working

The circuit diagram of the emergency light cum advertising display board is shown in Fig. 2. It comprises transformer X1, 12V voltage regulator LM7812 (IC1), timer NE555 (IC2), NPN transistor SL100 (T1), a 12V SPDT relay, and some other components. The transformer and bridge rectifier recharge the battery directly when mains 230V AC is connected across CON1.

In this circuit, the timer IC NE555 is configured in free-running multivibrator mode with its on and off period set at about 25 seconds. The pulse generated by timer IC NE555 at its pin 3 is fed to the base of NPN transistor SL100 via resistor R3. This pulse signal causes the transistor to start conducting, thereby energising the relay. Due to the magnetic properties of the relay, it attracts the attached plunger. As a result, the relay switches from normally open (N/O) mode to normally closed (N/C) mode. In both N/O and N/C modes of the relay, the LED strip available in the market, arranged in the form of alphabets 'V' and 'J,' also illuminates the nearby area.

After switch S1 is turned on, the first LED strip (alphabet 'V') lights up. After 25 seconds, the other LED strip (alphabet 'J') lights up. At this time, however, the LED strip alpha-

Fig. 1: Author's prototype

PARTS LIST

Semiconductors:	
IC1	- LM7812, 12V voltage regulator
IC2	- NE555 timer
T1	- SL100 NPN transistor
BR1	- 1A bridge rectifier
D1-D2	- 1N4007 rectifier diode
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 1-kilo-ohm
R2	- 390-kilo-ohm
R3	- 15-kilo-ohm
Capacitors:	
C1	- 1000µF, 35V electrolytic
C2	- 1µF, 35V electrolytic
C3	- 100µF, 35V electrolytic
C4	- 0.01µF, ceramic disk
Miscellaneous:	
BATT.1	- 12V, 1.3Ah battery
S1	- On/off switch
CON1-CON4	- 2-pin connectors
RL1	- 12V SPDT relay
X1	- 230V AC primary to 15V, 1A secondary transformer
	- 12V white LED strip (two)

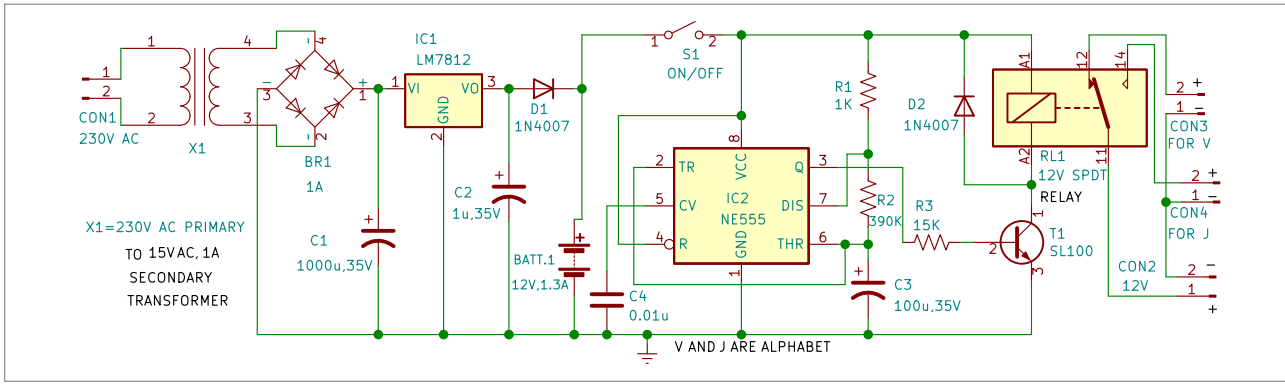

Fig. 2: Circuit diagram